

A 10-b 120-Msample/s Time-Interleaved Analog-to-Digital Converter With Digital Background Calibration

Shafiq M. Jamal, *Member, IEEE*, Daihong Fu, Nick C.-J. Chang, *Student Member, IEEE*, Paul J. Hurst, *Fellow, IEEE*, and Stephen H. Lewis, *Fellow, IEEE*

Abstract—Digital calibration using adaptive signal processing corrects for offset mismatch, gain mismatch, and sample-time error between time-interleaved channels in a 10-b 120-Msample/s pipelined analog-to-digital converter (ADC). Offset mismatch between channels is overcome with a random chopper-based offset calibration. Gain mismatch and sample-time error are overcome with correlation-based algorithms, which drive the correlation between a signal and its chopped image or its chopped and delayed image to zero. Test results show that, with a 0.99-MHz sinusoidal input, the ADC achieves a peak signal-to-noise-and-distortion ratio (SNDR) of 56.8 dB, a peak integral nonlinearity of 0.88 least significant bit (LSB), and a peak differential nonlinearity of 0.44 LSB. For a 39.9-MHz sinusoidal input, the ADC achieves a peak SNDR of 50.2 dB. The active area is 5.2 mm², and the power dissipation is 234 mW from a 3.3-V supply.

Index Terms—Adaptive systems, analog-to-digital conversion, calibration, CMOS analog integrated circuits.

I. INTRODUCTION

DIGITAL SIGNAL processing (DSP) systems operating on analog inputs are often limited by the sampling rates of analog-to-digital converters (ADCs). Time interleaving more than one ADC is a well-known technique used to increase the maximum sample rate [1]–[15]. Unfortunately, the performance of time-interleaved ADCs is sensitive to offset and gain mismatches as well as aperture errors between the interleaved channels. Digital foreground calibration can remove offset mismatch [7], [15] but requires interrupting the conversion of the input. To avoid such interruptions, analog background calibration has been used to remove offset and gain mismatch [10], but the required analog calibration circuits are not easily portable to evolving scaled technologies. Digital background calibration of offset and gain mismatch allows portability but was limited in several ways that are described in Section III [11]. Furthermore, to avoid the problem of aperture errors in previous work, a single front-rank sample-and-hold amplifier

(SHA) was used in front of all the interleaved channels [2], [9]–[12], [15], but a front-rank SHA limits the overall speed and therefore the number of channels that can be interleaved in practice. Digital calibration of sample-time errors has been proposed to overcome this limitation [13]; however, the proposed technique requires an accurate ramp signal to be applied to measure the timing errors.

To solve all these problems, this paper presents a time-interleaved ADC that uses digital-background calibration to overcome offset, gain, and sample-time errors between channels. The proposed structure does not use a front-rank SHA. The remainder of this paper is divided into five sections. Section II gives a brief review of time-interleaved ADCs and their limitations. Section III shows how the mismatch problems can be overcome by using digital background calibration. In Section IV, the implementation of the prototype is described. Section V presents the experimental results, and, finally, Section VI gives the conclusion.

II. TIME-INTERLEAVED ADC ARCHITECTURE

Fig. 1 shows a simplified block diagram of a time-interleaved ADC. It consists of M ADCs in parallel, an analog demultiplexer at the input, and a digital multiplexer at the output. Each channel operates at the overall sampling rate f_s divided by M . During operation, the analog demultiplexer selects each channel in turn to process the analog input signal. The corresponding digital multiplexer selects the digital output of the selected channel and forms an effectively high-speed ADC. With interleaving, the overall sampling rate is f_s . An advantage of this structure is that the overall sampling rate increases by a factor of M without using a new process technology. However, the performance of time-interleaved ADCs is sensitive to mismatches among the channels [1]–[15].

Offset mismatches among the ADC channels contribute to a periodic additive pattern in the output of the ADC array. In the frequency domain, this pattern appears as tones at integer multiples of the channel sampling rate f_s/M . Gain mismatches among the parallel channels cause amplitude modulation of the input samples by the sequence of channel gains. In the frequency domain, this error causes copies of the input signal spectrum to appear centered around integer multiples of the channel sampling rate. Ideally, each channel should sample T seconds after the previous channel, where $T = 1/f_s$. Deviations from the ideal sampling instants can be represented as a sequence

Manuscript received March 31, 2002; revised August 1, 2002. This work was supported by UC MICRO under Grant 01-050 and by the National Science Foundation under Grant CCR-99001925.

S. M. Jamal was with the University of California, Davis, CA 95616 USA. He is now with Marvell Semiconductor, Sunnyvale, CA 94086 USA.

D. Fu was with the University of California, Davis, CA 95616 USA. She is now with Maxim Integrated Products, Sunnyvale, CA 94086 USA.

N. C.-J. Chang, P. J. Hurst, and S. H. Lewis are with the Solid-State Circuits Research Laboratory, Department of Electrical and Computer Engineering, University of California, Davis, CA 95616 USA (e-mail: lewis@ece.ucdavis.edu).

Digital Object Identifier 10.1109/JSSC.2002.804327

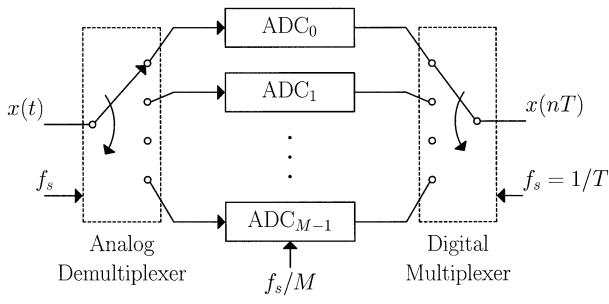
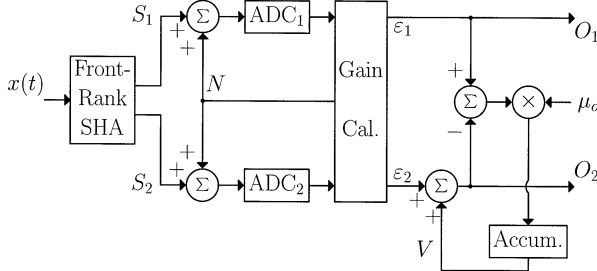


Fig. 1. Block diagram of the time-interleaved ADC architecture.

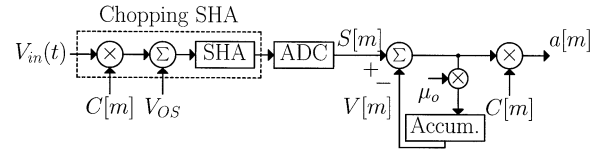
Fig. 2. Calibration system described by Fu *et al.* [11].

of sample-time errors that introduce errors in the input samples. The input samples are phase modulated by the sequence of sample-time errors in the ADC channels. In the frequency domain, this error produces copies of the input signal spectrum at the same frequencies as the spurious components stemming from gain mismatch. All of these mismatches cause the noise floor of the ADC to increase, thus reducing the system signal-to-noise ratio (SNR) [1]–[15].

III. DIGITAL BACKGROUND CALIBRATION

One method of doing background calibration in a time-interleaved ADC is to calibrate for the mismatches in the digital domain [11], as shown in Fig. 2. In this scheme, an analog calibration signal N is added to the input and the same value is digitally subtracted from the output to measure the gain mismatch. Then, one of the interleaved channels is multiplied by the appropriate gain to correct for the gain mismatch. To calibrate for offset mismatch, the average of the difference of the channel-one and channel-two outputs is added to channel-two. In the steady state, the offset of the channel-two output O_2 will equal the offset of the channel-one output O_1 . There are two disadvantages of these calibration methods. First, the calibration signal added to the ADC input reduces the dynamic range of the ADC. Second, the offset-calibration system introduces notches in the ADC output spectrum at integer multiples of the channel sample rate. Furthermore, in Fig. 2, a front-rank SHA is used to avoid sampling time errors [2]; however, this SHA limits the overall speed and therefore also the number of channels that can be interleaved in practice.

To eliminate the need for a calibration signal, background calibration of gain and sampling time errors is done here using the input signal itself. To eliminate the notches in the ADC output spectrum, a random-chopper-based offset calibration scheme removes offset mismatches. Also, a front-rank SHA is not used.

Fig. 3. Chopper-based offset calibration system for one channel. (V_{OS} represents the input-referred offset from the SHA and the ADC combined.)

Instead, the sampling operation is distributed among the channels by an array of SHAs (one per channel), and digital calibration is used to remove sampling-time errors.

A. Random Chopper-Based Offset Calibration

Fig. 3 shows a block diagram of a single channel that uses chopper-based offset calibration. It is similar to an independently developed technique [16]. A chopping SHA is used to multiply the analog input signal $V_{in}(t)$ by a pseudorandom binary signal $C[m] = \pm 1$, where m is a discrete-time index. $C[m]$ is white, has zero mean, and is uncorrelated with the input signal. Together with V_{OS} (which models the combined input-referred offset of the SHA and ADC), the chopped analog signal is then sampled and digitized by the ADC, producing signal $S[m]$. For simplicity, assume that the delay through the SHA and ADC is zero. A variable offset $V[m]$ is subtracted from $S[m]$, and the result is multiplied again by $C[m]$ to produce the channel output $a[m]$. Conceptually, the part of the channel output that is related to the analog input V_{in} is not affected by the chopping because it is multiplied by $C[m]$ twice and $C^2[m] = 1$.

Since the analog input has been converted to a white signal by the random chopper at the input, the chopped input for each channel contains little information at dc. Therefore, the dc component in the accumulator input stems mainly from the differences between the analog offset from the SHA and ADC in the channel and the accumulator output $V[m]$. In the steady state, feedback forces the average of the accumulator input to be zero. As a result, the offset of the channel stemming from both SHA and ADC offset is canceled in steady state by effectively introducing a notch at dc before the second chopper in Fig. 3. The scale factor μ_o controls the bandwidth of the notch, the speed of convergence, and the variance of $V[m]$ at convergence [17].

The offset calibration method in Fig. 3 can be extended to more than one channel by applying the same technique to each channel. Since this extension removes the offset of each channel independently, the same pseudorandom number $C[m]$ can be used for all the channels. In the steady state, the channel outputs are the digitized input samples with the offsets removed.

B. Gain Calibration

Fig. 4(a) shows the output spectrum of a two-channel time-interleaved ADC with a sinusoidal input at f_o and gain mismatch between the channels. Gain mismatch causes amplitude modulation, generating an image at f_i , which equals the channel sample rate $f_s/2$ minus f_o . The image amplitude is proportional to the gain mismatch $\Delta G = G_1 - G_2$, where G_1 is the gain of the top ADC channel and G_2 is the gain of the bottom ADC channel [18]. The concept of the gain calibration here is to use

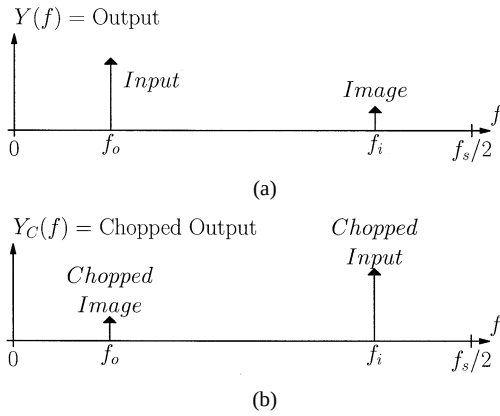


Fig. 4. ADC output spectrum for two time-interleaved channels with (a) a sinusoidal input at f_o and gain mismatch between the channels and (b) spectrum of chopped output.

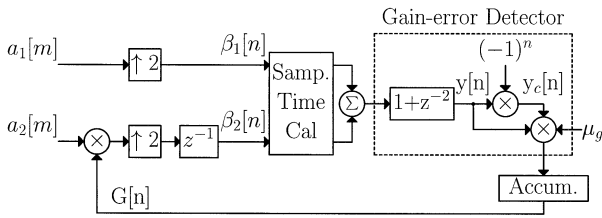


Fig. 5. Block diagram of the gain-calibration scheme.

this relationship between the input at f_o and the image at f_i to correct for the gain mismatch. First, the ADC output is chopped by multiplying it by a signal that alternates at the channel sampling rate. This multiplication causes the image to shift to f_o and the input to shift to f_i as shown in Fig. 4(b). Second, the output and the chopped output signals are multiplied in the time domain. The result has a dc component that is proportional to the gain mismatch.

Fig. 5 shows a block diagram of this gain-calibration scheme. $a_1[m]$ and $a_2[m]$ are the outputs of the two ADC channels after offset calibration. $a_1[m]$ is upsampled by a factor of two to produce $\beta_1[n]$ by inserting zero samples to produce a signal at the ADC sample rate of f_s . $a_2[m]$ is multiplied by a variable gain $G[n]$, upsampled by a factor of two, and delayed by one sample to produce $\beta_2[n]$. Because of this delay, the zero samples inserted in each path do not overlap (i.e., $\beta_1[n]\beta_2[n] = 0$). $\beta_1[n]$ and $\beta_2[n]$ are the inputs to the sample-time calibration block. The sample-time calibration block is described in the next section, but for now assume that this block just passes each signal through to the summer. The summer output goes into a gain-error detector.

At the input of the detector, the signal is passed through a short finite-impulse response (FIR) filter ($1 + z^{-2}$). Ignore this filter at first. The output of the FIR filter is $y[n]$, which is chopped to produce $y_c[n]$. Then, $y[n]$ and $y_c[n]$ are multiplied. The image at $f_s/2 - f_o$ turns out to be in phase with the input at f_o because the image is proportional to the input but insensitive to the slope of the input. Therefore, multiplying the chopped signal $y_c[n]$ by the unchopped signal $y[n]$ produces a signal with a dc component that is proportional to the gain mismatch ΔG . A small number μ_g scales the product of $y[n]$ and $y_c[n]$ to produce the accumulator input. The scale factor

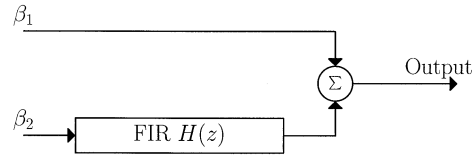


Fig. 6. Block diagram of the digital sample-time error-correction scheme.

μ_g controls the rate of convergence and the variance of $G[n]$ in steady state. Since each channel is upsampled by inserting zero samples, $y[n] = a_1[m]$ when n is even (i.e., $n = 2m$) and $y[n] = G[n] \cdot a_2[m]$ when n is odd (i.e., $n = 2m + 1$). When n is even, $(-1)^n = +1$ and $y_c[n] = y[n]$. In this case, $y_c[n] = a_1[m]$. However, when n is odd, $(-1)^n = -1$ and $y_c[n] = -y[n]$. In this case, $y_c[n] = -G[n] \cdot a_2[m]$. Since the feedback controls the gain for the bottom ADC channel and $G > 0$, the feedback in Fig. 5 is negative, and the average accumulator input converges to zero in the steady state. When this occurs, the correlation between the input and its chopped image is zero, and $G[n]$ has been adjusted to eliminate the gain mismatch between the two channels. $G[n]$ adjusts to be the ratio of G_1 to G_2 , which makes the gain of each channel from the analog input to β_1 or β_2 equal to G_1 .

A problem with this technique occurs when the ADC input has a frequency component at $f_s/4$. In this case, the product of $y[n]$ and $y_c[n]$ may generate a dc input to the accumulator even without gain mismatch between the two channels. [This occurs, for example, when $y[n] = \sin(n\pi/2)$.] This dc value would cause the variable gain $G[n]$ to adjust even if no adjustment is necessary. To overcome this problem, the filter $1 + z^{-2}$ produces a null at $f_s/4$ and prevents the loop from operating on inputs at this frequency (or inputs at frequencies that alias to $f_s/4$). For frequencies close to but not exactly equal to $f_s/4$, the loop gain and update rate are reduced because of the attenuation introduced by the $1 + z^{-2}$ filter. A sharper notch filter can be used if necessary.

Note that this gain-calibration algorithm does not require the injection of a calibration signal. Instead, the input signal itself is used for calibration. Signals at any frequency except $f_s/4$ (or frequencies that alias to $f_s/4$) contribute to the gain calibration. The rate of convergence is proportional to the input amplitude because the dc component at the accumulator input, which causes the gain adjustment, is proportional to the input amplitude. Therefore, the convergence is fast for large-amplitude inputs and slow for small-amplitude inputs, which is desirable because gain-error measurements are most accurate with maximum inputs.

C. Sample-Time Calibration

To correct sample-time errors, a digital sample-time-correction scheme is used. Fig. 6 shows a block diagram of the basic concept. β_1 and β_2 are the digital outputs of the gain-calibration system shown in Fig. 5. A digital FIR filter is inserted in the path of one of the channels. When the sampling clocks are ideal, this filter does nothing to the signal in the bottom channel. If a sample-time error exists, however, the FIR filter is configured to correct this error. The key idea is most easily explained by considering a simple case, where V_{in} is a sinusoid with frequency

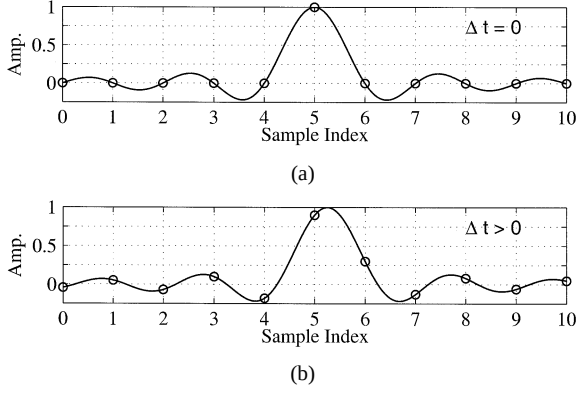


Fig. 7. Impulse response of the fractional-delay FIR filter (a) for no timing error ($\Delta t = 0$) and (b) with a timing error ($\Delta t > 0$).

$f_o < f_s/4$, and its image at $f_i = f_s/2 - f_o$ in β_1 and β_2 has been eliminated by digital low-pass filters with bandwidth $f_s/4$. If the ADC in the bottom channel samples $T + \Delta t$ seconds (rather than T seconds) after the top channel, a sample-time error occurs in the bottom channel. To correct this error, the FIR filter $H(z)$ introduces a fractional-sample-time delay in the bottom channel to calculate the sample values that would have occurred if the sample times had been correct. In other words, the FIR filter introduces a time delay that is a fraction of a sample period: $H(z) = z^{-\Delta t/T}$. [$H(z)$ also introduces a fixed delay that is an integer multiple of a sample period, but this fixed delay is ignored here for simplicity.] The sample-time error, Δt , can be positive or negative. With V_{in} band-limited to $f_s/4$, interpolation from one set of uniformly spaced samples to another in the bottom channel is possible because the sample rate in the lower channel is $f_s/2$. Therefore, no aliasing occurs during sampling and the continuous-time input could be regenerated using an ideal low-pass filter in the lower channel. Then the interpolated output could be produced by resampling the original continuous-time input. More simply, generation of one set of samples from another can be done using a discrete-time interpolation filter that convolves the samples $\beta_2[n]$ with a sampled sinc function.

To implement the fractional delay in practice, a causal FIR filter can be used [19], which introduces a delay of many sample periods in the lower signal path. Therefore, a delay must be added to the upper signal path to compensate for the delay through the FIR filter in the lower channel. Fig. 7 shows two plots of example 11-tap causal impulse responses. The filter coefficients are samples of a sinc function and the circles designate the values of the filter coefficients. When no sample-time error between the interleaved channels exists, $\Delta t = 0$ and all of the filter coefficients are zero except the center tap, which is one, as shown in Fig. 7(a). Therefore, the filter simply delays its input signal by an integer number of samples. However, when a timing error exists as shown in Fig. 7(b), all of the filter coefficients will have nonzero values and the filter delays its input signal by a noninteger number of samples.

The approach described above is valid as long as the continuous-time input signal V_{in} is band-limited to $f_s/4$ and β_1 and β_2 are low-pass filtered to eliminate the image at f_i due to sample-time error, which would appear between $f_s/4$ and $f_s/2$.

In this case, the FIR filter needs only to correct the phase shift experienced by the input signal due to the sample-time error in the bottom channel. Therefore, this approach is valid for input frequencies up to $f_s/4$. In practice, however, a preferable goal is to be able to handle input frequencies up to at least $f_s/2$. This goal can be achieved by using an all-pass filter $H(z)$ in the bottom channel that provides the appropriate phase shift to the signal β_2 so that the image is eliminated in the combined output of the two channels. Such a filter has the following frequency response [20]:

$$H(\omega) = e^{-j\omega\Delta t} e^{j(\Delta t/T)\pi \text{sign}(\omega)}, \quad -\omega_s/2 \leq \omega \leq \omega_s/2 \quad (1)$$

where Δt is the timing error and $T = 1/f_s$ is the sampling period of the interleaved ADC. This filter has a magnitude response of unity. Also, the negative of the slope of the phase response or the group delay of the filter is Δt except for discontinuities at kf_s , where k is any integer. This filter causes cancellation of the image caused by sampling time error for any input frequency between 0 and $f_s/2$ but introduces a small attenuation $\cos[\pi(\Delta t/2T)]$ and a constant phase shift $\pi(\Delta t/2T)$ in the combined output of the two channels. For $\Delta t/T < 1\%$, the resulting attenuation will be less than 1.1×10^{-3} dB and the phase shift will be less than 0.9° . However, both of these effects can be removed through the use of a second filter that processes the combined output if necessary.

The impulse response corresponding to the frequency response shown in (1) is

$$h[n] = -\frac{\sin(\pi \frac{\Delta t}{T} n)}{\pi(n - \frac{\Delta t}{T})} \quad (2)$$

where n is the filter coefficient index. An infinite number of filter taps is needed to perfectly compensate for the sample-time error, making the exact implementation of the filter impossible. In practice, truncating the number of filter taps is necessary. Accurate approximation of the transfer function in (1) can require a large number of filter taps. However, using windowing and least-mean squared error filter design [22], the number of taps can be optimized such that it is sufficient for the ADC requirements [22], [23].

Based on simulations, a 21-tap FIR filter with 10-b coefficients is sufficient to correct any timing error between ± 200 ps to 10-b accuracy for frequencies as high as 54 MHz, which is 90% of $f_s/2$ in the prototype. More taps are needed for larger timing errors or higher frequencies. Also, the FIR filter must be adaptive because the sample-time error is unknown.

Fig. 8 shows a block diagram of the adaptive sample-time calibration scheme. β_1 and β_2 are outputs of the gain-calibration system. β_1 goes through a fixed delay that equals the delay through the adaptive FIR filter when $\Delta t = 0$. β_2 goes through an adaptive FIR filter. The results from both channels are summed, and the summed output goes into a phase detector. The phase detector and accumulator here are the same as the detector in the gain-error calibration loop in Fig. 5 with one exception. The exception is that a delay of z^{-1} is introduced after the chopper. Ignoring this delay, this detector is the same as before because the image that arises from phase modulation appears at the same frequency as the image that arises from amplitude modulation. Therefore, almost the same method of calculating the correlation between the input and the image can be used here.

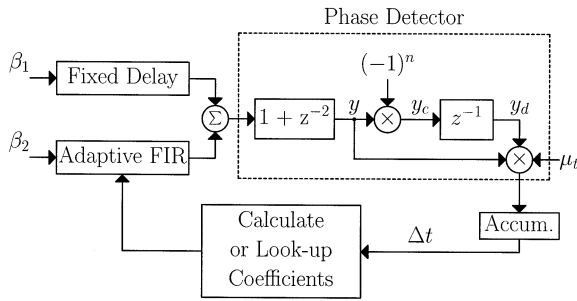


Fig. 8. Block diagram of the adaptive sampling-time calibration system.

However, the image at $f_s/2 - f_o$ is 90° out of phase with the input here, unlike in the previous case [20]. Physically, this orthogonality occurs because the image arising from timing error is proportional to the slope of the input signal, unlike in the case of gain mismatch between the channels. If the input is a superposition of sine waves with various amplitudes, phase shifts, and frequencies, the images stemming from timing error are proportional to the cosines of the corresponding inputs. Therefore, each image component is 90° out of phase with its corresponding input component. If the chopped signal $y_c[n]$ is multiplied directly by the unchopped signal $y[n]$, the result averages to zero because of this phase shift. An ideal solution to this problem would phase shift all positive frequency components of the chopped signal by $-\pi/2$ radians and multiply the phase-shifted, chopped signal by the unchopped signal. This solution would not only eliminate the orthogonality of the chopped timing images, but also would cause phase-shifted image components stemming from gain mismatch, after chopping, to become orthogonal to the corresponding components in the unchopped signal. As a result, this solution allows the timing error to be detected while a gain mismatch is present. Such a filter implements a discrete-time Hilbert transform [24].

The duration of the impulse response of such a filter is infinite, and truncation is required for implementation. Fig. 8 uses the simplest approximation to this filter, which is just a delay of one sample. This approximation gives unity magnitude response, as does the ideal Hilbert filter. Unlike the Hilbert filter, however, this approximation gives a phase shift that is a linear function of frequency: the phase shift is zero at zero frequency and $-\pi$ radians at $f_s/2$. As a result, the phase shift is exactly $-\pi/2$ radians only at $f_s/4$, but signals at this frequency are blocked by the $1 + z^{-2}$ filter at the input of the phase detector. Therefore, with a unit-delay approximation to the Hilbert filter, the phase shift is not exactly $-\pi/2$ radians at any frequency that contributes to the timing-error detection. For example, for very low frequencies, the phase shift is almost zero, and the chopped image components remain nearly orthogonal to their corresponding input components. However, timing errors in this case are insignificant because the slope of the input signal is small. Similarly, for frequencies near $f_s/2$, the phase shift introduced by z^{-1} is nearly $-\pi$ radians, and the corresponding image components, after chopping, remain nearly orthogonal to the input. For frequencies above about 90% of $f_s/2$, however, the chosen interpolation filter cannot provide adequate interpolation for 10-b accuracy. Therefore, the fact that the delay of z^{-1}

provides a phase shift of nearly $-\pi$ radians instead of $-\pi/2$ radians for these frequencies is of little consequence here.

With the z^{-1} approximation to the Hilbert filter, most of the detection of the timing error is determined by inputs near $f_s/4$ but far enough away from this frequency that the attenuation introduced by the $1 + z^{-2}$ filter is insignificant. Also, with this approximation, the phase detector in Fig. 8 not only senses timing error but also gain mismatch. This is because the phase shift is not exactly $-\pi/2$ radians for any frequency that contributes to the timing detection. As a result, chopped image components stemming from gain mismatch are not orthogonal to their corresponding input components in Fig. 8. Fortunately, the chopped image components arising from timing errors remain orthogonal to the corresponding input components in the gain-calibration loop in Fig. 5. Therefore, the gain-error detector correctly senses the gain mismatch even in the presence of timing error. After the gain-calibration loop in Fig. 5 converges, the timing loop in Fig. 8 converges to find the correct timing error.

In some cases, more accurate approximations to the adaptive interpolation filter and/or the Hilbert filter than used here may be required. For example, if the input signal only contains components near 95% of $f_s/2$, more than 21 taps in the interpolating filter would be required for 10-b accuracy. Also, in this case, a longer accumulator and/or more than a one-tap approximation to the Hilbert filter may be required to sense timing errors. Furthermore, in other cases, the filter that provides a notch at $f_s/4$ might have to be sharpened. For example, if the input signal only contains components near $f_s/4$ where the attenuation introduced by the $1 + z^{-2}$ filter is too severe to allow correct operation of the phase detector, a sharper notch filter would be required. A key point here is that the structure proposed above can correctly calibrate in the background for many inputs in principle, but knowledge of the input characteristics is helpful in designing an efficient calibration system in practice.

Finally, note that the frequency responses of the digital filters mentioned above are periodic with period f_s , but a sinusoidal input experiences a phase shift due to sample-time error that is periodic with period $1/\Delta t$. Therefore, the sample-time correction using $H(z)$ in (1) works only for input frequencies below $f_s/2$. However, operation for input frequencies above $f_s/2$ is possible by changing $H(z)$ to provide the proper phase shift to eliminate the resulting image. To correct for sample-time errors for inputs band limited from $kf_s/2$ to $(k+1)f_s/2$, the ideal $H(z)$ differs for each k , where k is any positive integer. The calibration can work for any band from $kf_s/2$ to $(k+1)f_s/2$ as long as the value of k is known in advance.

The only other difference from the gain-calibration case is that the accumulator output here does not control the gain of a multiplier in the bottom channel. Instead, the accumulator output is a measure of the timing error Δt that is used to calculate or look up the required FIR filter coefficients.

IV. PROTOTYPE IMPLEMENTATION

To demonstrate the capabilities and limitations of the digital background calibration techniques, a prototype with two ADCs in parallel was designed and fabricated in a $0.35\text{-}\mu\text{m}$ CMOS technology. Fig. 9 shows the block diagram of one

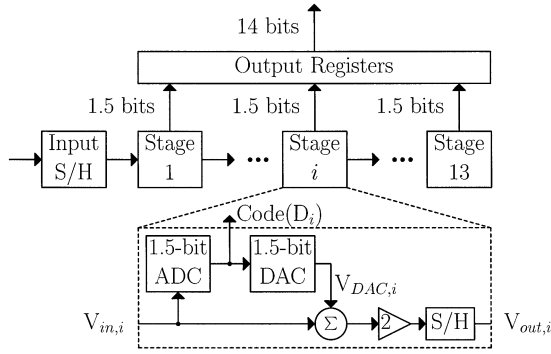


Fig. 9. Block diagram of the pipelined ADC in each channel. The resolution is 14 b, but testing shows that 12-b resolution is adequate.

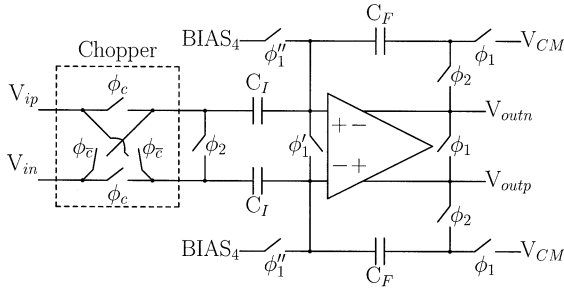


Fig. 10. Chopping SHA (if $C[m] = 1$, $\phi_c = \phi_1$, and $\phi_\varepsilon = 0$; if $C[m] = -1$, $\phi_c = 0$, and $\phi_\varepsilon = \phi_1$).

pipelined ADC. A 1.5-b/stage architecture is used. The goal was to achieve 10-b performance in the parallel ADC. Two 14-b pipelined ADCs were used in the prototype, so each channel has 13 1.5-b stages. The resolutions of the ADCs were increased from 10 b to allow testing with more than ten digital outputs. However, testing reveals that a 10-b output in each ADC is adequate. (For example, using a 12-b output in each channel and then truncating the final output to 10 b after calibration improves the performance by at most 1 dB.)

Fig. 10 shows the chopping SHA used in the prototype. It is a bottom-plate SHA with additional switches at the input for chopping. The circuit uses conventional nonoverlapping clocks with two primary phases ϕ_1 and ϕ_2 along with two extra phases ϕ'_1 and ϕ'_2 to reduce the sample-to-hold transition error [25]. When the chopping signal $C[m] = 1$ during the sampling phase, the top and bottom switches are on, and V_{ip} is connected to the top input capacitor and V_{in} is connected to the bottom input capacitor. When the chopping signal $C[m] = -1$, the inputs are crossed so V_{ip} is connected to the bottom input capacitor and V_{in} is connected to the top input capacitor.

Fig. 11 shows the schematic of the operational amplifier (opamp). A telescopic structure was used to minimize settling time [26], and source followers were used at the input to reduce the input capacitance and increase the feedback factor in a closed-loop configuration [27]. If all of the stages in the ADC are identical, the loop gain in each stage should be more than 2^{10} or 60 dB to keep the peak integral and differential nonlinearities less than 0.5 LSB at a 10-b level. If the feedback factor is $1/3$, the required open-loop gain in the opamp is about 70 dB [18], [28]. Also, to reach a conversion rate of 60 Msample/s for each channel, the opamps must settle within

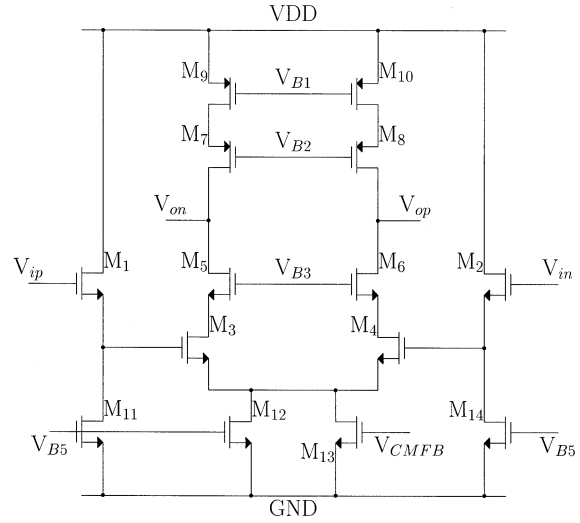


Fig. 11. Opamp schematic.

0.05% accuracy in less than 8.3 ns. Therefore, the opamp needs both high gain and bandwidth. The opamp used in the prototype has an open-loop gain of about 50 dB and a settling time of about 7 ns. With 1 ns of nonoverlap time and 1 ns of total rise and fall times, the minimum conversion period for each channel is limited to about 16 ns, allowing each channel to operate up to about 62.5 Msample/s. The loop gain of the opamp normally would be enough for about 7- or 8-b linearity. The linearity is improved here by adjusting the reference voltages of the first three pipeline stages independently to overcome the effects of interstage gain errors. These adjustments can be made automatically using background calibration [29]. To simplify the prototype, however, here the adjustments were made by hand.

To minimize die area and power dissipation, the capacitors in a pipelined ADC can be scaled in every stage [30], [31]. However, this approach increases the design time to optimize each stage. To reduce the design time, the capacitors are scaled only once in this prototype. In the first three stages, the capacitance of each sampling capacitor is 0.5 pF. In the later stages, this capacitance is 0.125 pF.

An important circuit cell in the pipelined ADC is the comparator. Two types of comparators are used in the prototype. In the first three stages, a comparator with a preamp and the same schematic as in [32] is used. In the other stages, a dynamic comparator with built-in mismatch to set the threshold is used [30]. This comparator has zero dc power dissipation and does not require sampling capacitors. However, testing revealed that these comparators in the prototype have offsets of approximately 300 mV. The effect of the comparator offset on the performance of the prototype is small because the offset was within the correction range of the ADC; however, this comparator may not be useful with a reduced supply and correction range.

A key block in the offset calibration system is the pseudo-random-number generator (RNG). At a rate of $f_s/2$, it generates a binary sequence $C[m]$ that is white and uncorrelated with the input signal. The RNG is implemented on the chip and uses a maximum-length shift-register structure with 41 stages in the shift register [33]. At a sample rate of 60 Msample/s, it yields a

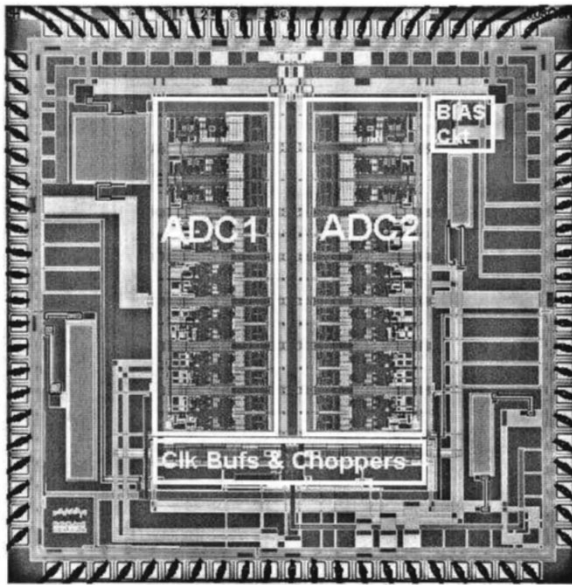


Fig. 12. Die photograph.

period of about 10 h, so that the periodicity of the noise is well below any measurement frequency of interest.

Fig. 12 shows the die photo of the prototype. It is fabricated in a $0.35\text{-}\mu\text{m}$ CMOS technology that has poly-poly capacitors and four layers of metal. The dimensions of the prototype are $3.5\text{ mm} \times 3.5\text{ mm}$. The die area is 12.3 mm^2 , and the active area is about 5.2 mm^2 . The prototype consists of the two interleaved ADC channels but does not include the digital calibration circuits for simplicity. These calibration algorithms have been implemented in software using MATLAB and C programs as well as in a DSP chip. On-chip calibration circuits are estimated to require about 5 mm^2 and about 190 mW when running at the full sample rate in $0.35\text{-}\mu\text{m}$ CMOS. (The calibration loop can run at a slower rate, but the adaptive FIR filter in Fig. 8 must run at full speed and would consume about 90% of the calibration power.) With $0.15\text{-}\mu\text{m}$ CMOS, however, the area and power dissipation for the calibration circuit would scale to about 0.9 mm^2 and 40 mW , making the overhead small enough to consider using these techniques in products.

V. EXPERIMENTAL RESULTS

The following conditions were used for all tests described below unless stated otherwise. The power supply voltage is 3.3 V , and the temperature is 25°C . The input amplitude is $3\text{ V}_{\text{P-P}}$, and the input frequency is 0.99 MHz . The sampling and conversion rate is 120 Msample/s , but the ADC output was down-sampled by a factor of five so that our logic analyzer could collect the ADC output data. The calibration is performed in software on a workstation.

Fig. 13 shows the ADC output spectrum with and without calibration. The y axes in both plots in Fig. 13 are normalized so that the input amplitude is 0 dB . Fig. 13(a) shows the output of the ADC without calibration. The tone caused by offset mismatch is at the channel sampling rate of 60 MHz , which appears at 12 MHz after down sampling. The offset tone is 47.4 dB below the input. The tone caused by gain and timing mismatch is at the channel sampling rate minus 0.99 MHz , which appears

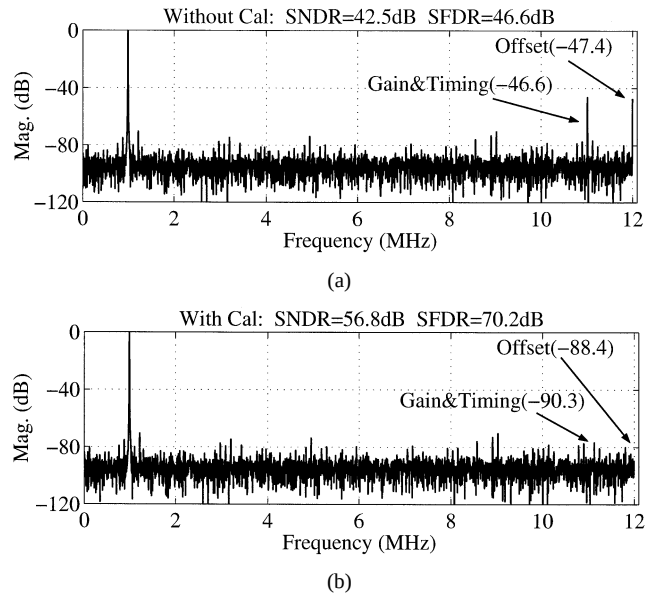


Fig. 13. ADC output spectrum (a) without calibration and (b) with calibration. ($f_s = 120\text{ Msample/s}$, $V_{\text{in}} = 3\text{ V}_{\text{P-P}}$, and $f_o = 0.99\text{ MHz}$).

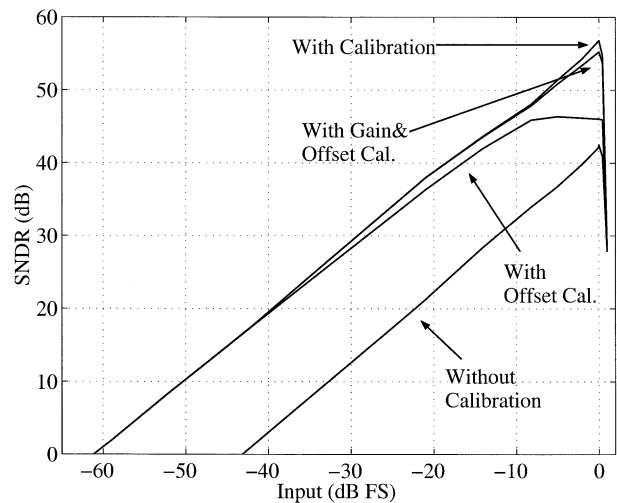


Fig. 14. SNDR versus input amplitude ($f_s = 120\text{ Msample/s}$, $f_o = 0.99\text{ MHz}$).

at 11.01 MHz after downsampling. This tone is 46.6 dB below the input. Although the gain and timing mismatch tones appear together, the main contribution here comes from the gain mismatch because the input frequency is low. Without calibration, the signal-to-noise-and-distortion ratio (SNDR) is 42.5 dB , and the spurious-free dynamic range (SFDR) is 46.6 dB . Fig. 13(b) shows the output of the ADC with calibration. The offset tone has decreased by 41 dB and is now 88.4 dB below the input. The gain and timing tones have decreased by 43.7 dB now to be 90.3 dB below the input. The overall SNDR here has improved by $14.3\text{--}56.8\text{ dB}$, and the SFDR has improved by $23.6\text{--}70.2\text{ dB}$.

Fig. 14 shows four plots of the SNDR versus input amplitude. The bottom plot is without calibration. The next is with offset calibration. The third plot is with gain and offset calibration. The top is with offset, gain, and timing calibration. Without calibration, the dynamic range is 43.2 dB , and the maximum SNDR is 42.5 dB . With offset calibration, the dynamic range improves by $18.3\text{--}61.5\text{ dB}$, and the maximum SNDR increases

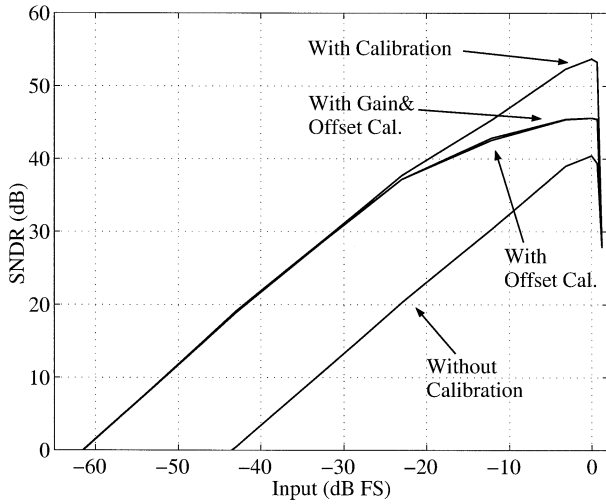


Fig. 15. SNDR versus input amplitude ($f_s = 120$ Msample/s, $f_o = 9.90$ MHz). The plot “With Gain and Offset Cal.” is almost identical to the plot “With Offset Cal.”

by 5.0–47.5 dB. Since offset mismatch is not proportional to the input, it creates the dominant error for small inputs. Adding gain calibration to offset calibration does not change the dynamic range because gain errors are insignificant for small inputs. However, gain errors are significant for large inputs, and adding gain calibration increases the maximum SNDR by approximately 7.9–55.4 dB. Finally, since the input frequency is low here (0.99 MHz), timing errors are small, and adding timing calibration only improves the SNDR by 1.4–56.8 dB.

Fig. 15 shows four plots of the SNDR versus input amplitude for the same cases as in Fig. 14 with one exception. In Fig. 15, the input frequency is 9.90 MHz, which is a factor of 10 higher than in Fig. 14. Without calibration, the dynamic range is 43.1 dB, and the maximum SNDR is 40.5 dB. With offset calibration, the dynamic range increases by 18.1–61.3 dB, and the maximum SNDR increases by 5.1 dB to about 45.6 dB. With gain and offset calibration, the dynamic range and maximum SNDR are the same as with offset calibration only. This is because the offset mismatch dominates for small inputs and the timing error dominates for large inputs in this case. With offset, gain, and timing calibration, the dynamic range stays at about 61.3 dB, but the maximum SNDR increases by 8.4 dB to approximately 53.9 dB.

Fig. 16 shows three plots of SNDR versus input frequency. The bottom plot is without calibration. The top plot is with offset, gain, and timing calibrations implemented in software on a workstation. The middle plot is with offset, gain, and timing calibrations implemented with a DSP chip.¹ To allow the DSP chip to process the data, the ADC output was down sampled by a factor of 375 for the data in the middle plot. This means that the ADC was operating at a sampling and conversion rate of 120 Msample/s, but new inputs were loaded into the DSP chip at a rate of 320 ksampling/s for the middle plot. With workstation-based offset, gain, and timing calibration, the SNDR is above 50 dB for input frequencies up to 40 MHz. The plot with DSP-based calibration is between about 1 and 3 dB lower than the software-based plot up to about 56 MHz. The calibration al-

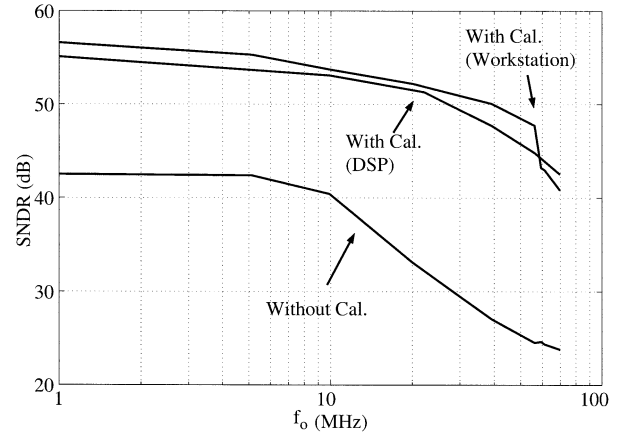


Fig. 16. SNDR versus input frequency f_o ($f_s = 120$ Msample/s).

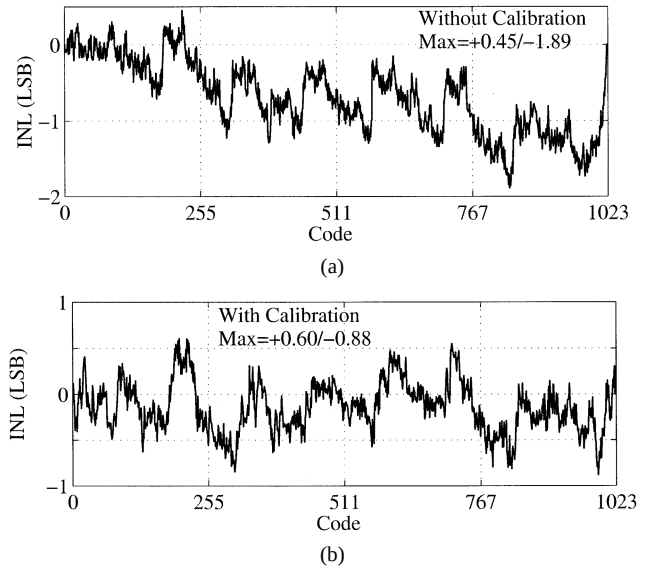


Fig. 17. INL versus code (a) without calibration and (b) with calibration ($f_s = 120$ Msample/s, $f_o = 0.99$ MHz).

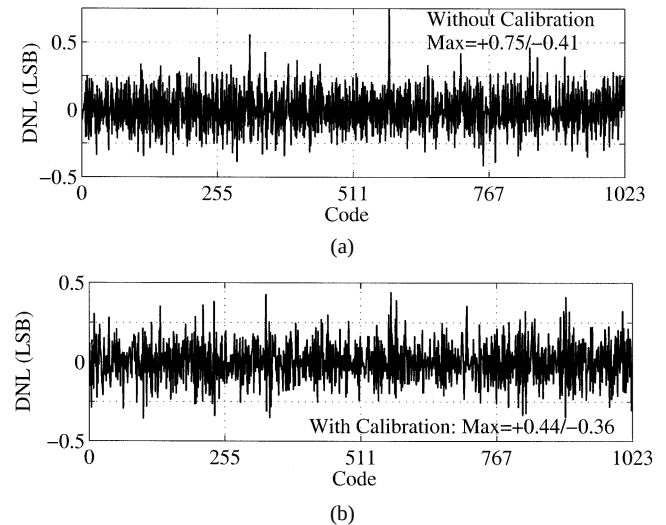


Fig. 18. DNL versus code (a) without calibration and (b) with calibration ($f_s = 120$ Msample/s, $f_o = 0.99$ MHz).

gorithms worked equally well in both cases, and the difference between the two plots with calibration stems from better perfor-

¹The DSP chip is a TMS320C6211 donated by Texas Instruments.

TABLE I
PERFORMANCE SUMMARY (3.3 V, 25 °C)

	Without Calibration	With Calibration
Process	0.35- μ m double-poly CMOS	0.35- μ m double-poly CMOS
Resolution	10 bits	10 bits
Sampling rate	120 Msample/s	120 Msample/s
Active Area	5.2 mm ²	5.2 mm ² + External
Power Dissipation (Analog/Total)	171 mW / 234 mW	171 mW / 234 mW + External
Full-Scale Input	3 V peak-to-peak	3 V peak-to-peak
$\mu_o = \mu_g = \mu_t$	0	2^{-22}
DNL ($f_o=0.99$ MHz)	+0.75/-0.41 LSB	+0.44/-0.36 LSB
INL ($f_o=0.99$ MHz)	+0.45/-1.89 LSB	+0.60/-0.88 LSB
THD ($f_o=0.99$ MHz)	-62.4 dB	-62.4 dB
SNDR ($f_o=0.99$ MHz)	42.5 dB	56.8 dB
SFDR ($f_o=0.99$ MHz)	46.6 dB	70.2 dB
PSRR ($f_{supply}=0.05$ MHz)	67.0 dB	67.0 dB
CMRR ($f_o=0.99$ MHz)	68.0 dB	68.0 dB
Dynamic Range ($f_o=0.99$ MHz)	43.1 dB	61.5 dB

mance of the individual channels in the workstation case than in the DSP case.

Fig. 17 shows the integral nonlinearity (INL) plots. Fig. 17(a) shows the plot of INL without calibration, and Fig. 17(b) shows the plot of INL with calibration. Without calibration, the peak INL is -1.89 least significant bit (LSB), and with calibration the peak INL decreases to -0.88 LSB. Fig. 18 shows the differential nonlinearity (DNL) plots. Fig. 18(a) shows the plot of DNL without calibration, and Fig. 18(b) shows the plot of DNL with calibration. Without calibration, the peak DNL is +0.75 LSB and with calibration the peak DNL decreases to +0.44 LSB. Neither the DNL nor the INL is expected to improve after calibration. The purpose of the calibration here is to make the channels appear to be identical in terms of their offsets, gains, and aperture errors. Table I summarizes the measured performance.

VI. CONCLUSION

Time-interleaving of monolithic ADCs was first described in a paper by Black and Hodges 22 years ago [1]. However, the use of time interleaving in ADC products has been limited since then mainly by offset, gain, and timing mismatches between the interleaved channels. This paper describes the first demonstration of digital-background-calibration techniques to overcome the effects of all three of these error sources. The presented system uses a random chopper-based algorithm for offset calibration and correlation-based algorithms for gain and sampling-time calibration. This system does not introduce any spectral nulls and uses the input signal to calibrate for gain mismatch and sampling-time errors. Since the calibration is done in the digital domain here, the overhead required in terms of area and power dissipation to implement monolithic background calibration is expected to scale dramatically in state-of-the-art process technologies. This work potentially expands the range of applications in which time-interleaving is a viable solution to reach the high conversion rates required in future ADC products.

ACKNOWLEDGMENT

The authors are grateful to K. C. Dyer and R. Stevens at Intel Corporation for doing a fixed-ion-beam repair on the prototype. They are also grateful to J. B. Fowler, R. Cox, and T. L. Viswanathan at Texas Instruments for their help in providing a DSP starter kit. Finally, the authors are grateful to J. Corcoran

at Agilent Technologies for helpful technical discussions at the beginning of this project.

REFERENCES

- [1] W. C. Black, Jr. and D. A. Hodges, "Time interleaved converter arrays," *IEEE J. Solid-State Circuits*, vol. SC-15, no. 6, pp. 1022-1029, Dec. 1980.
- [2] K. Poulton, J. J. Corcoran, and T. Hornak, "A 1-GHz 6-bit ADC system," *IEEE J. Solid-State Circuits*, vol. SC-22, no. 6, pp. 962-970, Dec. 1987.
- [3] Y. C. Jenq, "Digital spectra of nonuniformly sampled signals: Fundamentals and high-speed waveform digitizers," *IEEE Trans. Instrum. Meas.*, vol. 37, pp. 245-251, June 1988.
- [4] Y. C. Jenq, "Digital spectra of nonuniformly sampled signals: A robust sampling time offset estimation algorithm for ultra high-speed waveform digitizers using interleaving," *IEEE Trans. Instrum. Meas.*, vol. 39, pp. 71-75, Feb. 1990.
- [5] A. Petraglia and S. K. Mitra, "Analysis of mismatch effects among A/D converters in a time-interleaved waveform digitizer," *IEEE Trans. Instrum. Meas.*, vol. 40, pp. 831-835, Oct. 1991.
- [6] M. Yotsuyanagi, T. Etoh, and K. Hirata, "A 10-b 50-MHz pipelined CMOS A/D converter with S/H," *IEEE J. Solid-State Circuits*, vol. 28, pp. 292-300, Mar. 1993.
- [7] C. S. G. Conroy, D. W. Cline, and P. R. Gray, "An 8-b 85-MS/s parallel pipeline A/D converter in 1- μ m CMOS," *IEEE J. Solid-State Circuits*, vol. 28, pp. 447-454, Apr. 1993.
- [8] K. Nakamura, M. Hotta, L. R. Carley, and D. J. Allstot, "An 85 mW, 10b, 40 Msample/s CMOS parallel-pipelined ADC," *IEEE J. Solid-State Circuits*, vol. 30, pp. 173-183, Mar. 1995.
- [9] K. Y. Kim, N. Kusayanagi, and A. A. Abidi, "A 10-b, 100-MS/s CMOS A/D converter," *IEEE J. Solid-State Circuits*, vol. 32, pp. 302-311, Mar. 1997.
- [10] K. C. Dyer, D. Fu, S. H. Lewis, and P. J. Hurst, "An analog background calibration technique for time-interleaved analog-to-digital converters," *IEEE J. Solid-State Circuits*, vol. 33, pp. 1912-1919, Dec. 1998.
- [11] D. Fu, K. C. Dyer, S. H. Lewis, and P. J. Hurst, "A digital background calibration technique for time-interleaved analog-to-digital converters," *IEEE J. Solid-State Circuits*, vol. 33, pp. 1904-1911, Dec. 1998.
- [12] K. C. Dyer, D. Fu, S. H. Lewis, and P. J. Hurst, "A comparison of monolithic background calibration in two time-interleaved analog-to-digital converters," in *Proc. IEEE Int. Symp. Circuits and Systems*, vol. 1, May 1998, pp. 13-16.
- [13] H. Jin and E. K. F. Lee, "A digital-background calibration technique for minimizing timing-error effects in time-interleaved ADCs," *IEEE Trans. Circuits Syst. II*, vol. 47, no. 7, pp. 603-613, July 2000.
- [14] N. Kurosawa, H. Kobayashi, K. Maruyama, H. Sugawara, and K. Kobayashi, "Explicit analysis of channel mismatch effects in time-interleaved ADC systems," *IEEE Trans. Circuits Syst. I*, vol. 48, pp. 261-271, Mar. 2001.
- [15] L. Sumanen, M. Waltari, and K. A. I. Halonen, "A 10-bit 200-MS/s CMOS parallel pipeline A/D converter," *IEEE J. Solid-State Circuits*, pp. 1048-1055, July 2001.
- [16] H. van der Ploeg, G. Hoogzaad, H. A. H. Termmer, M. Vertregt, and R. L. J. Roovers, "A 2.5-V 12-b 54-Msample/s 0.25- μ m CMOS ADC in 1 mm² with mixed-signal chopping and calibration," *IEEE J. Solid-State Circuits*, vol. 36, no. 12, pp. 1859-1867, Dec. 2001.
- [17] B. Widrow and S. D. Stearns, *Adaptive Signal Processing*. Englewood Cliffs, NJ: Prentice-Hall, 1985.

- [18] C. S. G. Conroy, "A high-speed parallel pipelined A/D converter technique in CMOS," Ph.D. dissertation, Univ. California at Berkeley, Feb. 1994.
- [19] T. I. Laakso, V. Välimäki, M. Karjalainen, and U. K. Laine, "Splitting the unit delay, tools for fractional delay filter design," *IEEE Signal Processing Mag.*, vol. 13, no. 1, pp. 30–60, Jan. 1996.
- [20] S. M. Jamal, "Digital background calibration of time-interleaved analog-to-digital converter," Ph.D. dissertation, Univ. California at Davis, Sept. 2001.
- [21] A. V. Oppenheim and R. W. Schaffer, *Discrete-Time Signal Processing*. Englewood Cliffs, NJ: Prentice-Hall, 1989.
- [22] G. D. Cain, N. P. Murphy, and A. Tarczynski, "Evaluation of several variable FIR fractional-sample delay filters," in *IEEE Proc. Int. Conf. Acoust. Speech Signal Processing*, vol. 3, Apr. 1994, pp. 621–624.
- [23] C. S. Burrus, A. W. Sowieto, and R. A. Gopinath, "Least squared error FIR filter design with transition bands," *IEEE Trans. Signal Processing*, vol. 40, no. 6, pp. 1327–1340, June 1992.
- [24] E. A. Lee and D. G. Messerschmitt, *Digital Communication*, 2nd ed. Boston, MA: Kluwer, 1994, p. 273.
- [25] Y. M. Lin, B. Kim, and P. R. Gray, "A 13-b 2.5-MHz self-calibrated pipelined A/D converter in 3- μ m CMOS," *IEEE J. Solid-State Circuits*, vol. 26, pp. 628–636, Apr. 1991.
- [26] G. Nicollini, P. Confalonieri, and D. Senderowicz, "A fully differential sample-and-hold circuit for high-speed applications," *IEEE J. Solid-State Circuits*, vol. 24, no. 10, pp. 1461–1465, Oct. 1989.
- [27] W.-C. Song, H.-W. Choi, S.-U. Kwak, and B.-S. Song, "A 10-b 20-Msample/s low-power CMOS ADC," *IEEE J. Solid-State Circuits*, vol. 30, no. 5, pp. 514–521, May 1995.
- [28] S. H. Lewis, "Optimizing the stage resolution in pipelined, multistage, analog-to-digital converters for video-rate applications," *IEEE Trans. Circuits Syst. II*, vol. 39, no. 8, pp. 516–523, Aug. 1992.
- [29] J. Ming and S. H. Lewis, "An 8-bit 80-Msample/s pipelined analog-to-digital converter with background calibration," *IEEE J. Solid-State Circuits*, vol. 36, pp. 1489–1497, Oct. 2001.
- [30] T. B. Cho and P. R. Gray, "A 10 b, 20 Msample/s, 35 mW pipeline A/D converter," *IEEE J. Solid-State Circuits*, vol. 30, pp. 166–172, Mar. 1995.
- [31] D. W. Cline and P. R. Gray, "A power optimized 13-b 5 Msamples/s pipelined analog-to-digital converter in 1.2 μ m CMOS," *IEEE J. Solid-State Circuits*, vol. 31, no. 3, pp. 294–303, Mar. 1996.
- [32] K. Nagaraj, H. Fetterman, J. Anidjar, S. Lewis, and R. Renninger, "A 250-mW, 8-b, 52-Msamples/s parallel-pipelined A/D converter with reduced number of amplifiers," *IEEE J. Solid-State Circuits*, vol. 32, no. 3, pp. 312–320, Mar. 1997.
- [33] R. C. Dixon, *Spread Spectrum Systems*. New York: Wiley, 1976.



Shafiq M. Jamal (S'94–M'01) was born in Kabul, Afghanistan, in 1974. He received the B.S., M.S., and Ph.D. degrees, all in electrical engineering, from the University of California at Davis, in 1996, 1999, and 2001, respectively.

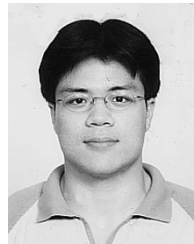
Since July 2001, he has been a Staff Design Engineer at Marvell Semiconductor Inc., Sunnyvale, CA. His current research interests include mixed-signal circuit design in data communication and wireless communication.



Daihong Fu received the B.S. degree in electrical engineering from Tsinghua University, China, in 1989, the M.S. degree in mathematics from Lamar University, TX, in 1992, and the M.S. and Ph.D. degrees in electrical engineering from the University of California at Davis in 1996 and 1998, respectively.

She has been with Maxim Integrated Products, Sunnyvale, CA, since 1998. She has worked on the design of various analog integrated circuits for audio, supervisory, and temperature sensor applications. Her research interests include analog

and mixed-signal circuit design.



Nick C.-J. Chang (S'01) received the B.S. and M.S. degrees in electrical engineering from Columbia University, New York, NY, in 1998 and 1999, respectively. He is currently working toward the Ph.D. degree at the University of California at Davis.

He was an intern at Lucent Technologies, Inc., Allentown, PA, during the summer of 1999. His research interests include mixed-signal circuit design and signal processing applications.



Paul J. Hurst (S'76–M'83–SM'94–F'01) received the B.S., M.S., and Ph.D. degrees in electrical engineering from the University of California at Berkeley in 1977, 1979, and 1983, respectively.

From 1983 to 1984, he was with the University of California at Berkeley as a lecturer, teaching integrated-circuit design courses and working on an MOS delta-sigma modulator. In 1984, he joined the telecommunications design group of Silicon Systems Inc., Nevada City, CA, where he was involved in the design of CMOS integrated circuits for voice-band modems. Since 1986, he has been on the faculty of the Department of Electrical and Computer Engineering at the University of California at Davis, where he is now a Professor. His research interests are in the areas of data converters and analog and mixed-signal integrated-circuit design for communication applications. He was a member of the program committee for the Symposium on VLSI Circuits in 1994 and 1995, a member of the program committee for the International Solid-State Circuits Conference from 1998 until 2001, and a Guest Editor for the December 1999 issue of the *Journal of Solid-State Circuits*. He is now an Associate Editor for the *Journal of Solid-State Circuits*. He is a coauthor of the fourth edition of the textbook *Analysis and Design of Analog Integrated Circuits* (New York: Wiley, 2001). He is also active as a consultant to industry.



Stephen H. Lewis (S'85–M'88–SM'97–F'01) received the B.S. degree from Rutgers University, New Brunswick, NJ, in 1979, the M.S. degree from Stanford University, Stanford, CA, in 1980, and the Ph.D. degree from the University of California at Berkeley (UC Berkeley) in 1987, all in electrical engineering.

From 1980 to 1982, he was with Bell Laboratories, Whippany, NJ, where he was involved in circuit design for magnetic recording. In 1988, he rejoined Bell Laboratories, Reading, PA, where he concentrated on the design of analog-to-digital converters. In 1991, he joined the Department of Electrical and Computer Engineering, University of California at Davis, where he is now a Professor. He is a coauthor of a college textbook on analog integrated circuits, and his research interests include data conversion, signal processing, and analog circuit design.

Dr. Lewis received the award for the Outstanding Engineering Scholar at Rutgers University, the Sakris Memorial Prize at UC Berkeley, and the IEEE Third Millennium Medal. Also, he was a co-recipient of the Jack Kilby Award for Outstanding Student Paper and the Beatrice Winner Award for Editorial Excellence at International Solid-State Circuits Conference (ISSCC). He was a member of the Program Committee for the ISSCC from 1994 to 1998, an Associate Editor of the *IEEE JOURNAL OF SOLID-STATE CIRCUITS* from 1994 to 1997, and Editor of the *IEEE JOURNAL OF SOLID-STATE CIRCUITS* from 1998 to 2001. He is now Vice President of the IEEE Solid-State Circuits Society.